

Meyer Section 5.6 Solutions: Unitary and Orthogonal Matrices

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Source section. Carl D. Meyer, *Matrix Analysis and Applied Linear Algebra*, Section 5.6, “Unitary and Orthogonal Matrices.”

Exercise range solved. Exercises 5.6.1 through 5.6.21.

Copyright note. The exercise prompts are restated only briefly. I include the mathematical data needed for each solution, but I do not reproduce the textbook problem statements verbatim.

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Key ideas from Section 5.6

A complex square matrix U is **unitary** precisely when its columns are an orthonormal basis, equivalently

$$U^*U = I, \quad U^{-1} = U^*, \quad \|Ux\| = \|x\| \quad \text{for all } x.$$

A real square matrix P is **orthogonal** precisely when

$$P^T P = I, \quad P^{-1} = P^T, \quad \|Px\| = \|x\| \quad \text{for all } x.$$

Thus unitary and orthogonal matrices are the matrix versions of isometries.

For a unit vector u , the elementary orthogonal projector onto $u^\perp$ is

$$Q = I - uu^*,$$

and the orthogonal projection of x onto $\text{span}\{u\}$ is uu^*x . If $u \neq 0$ is not normalized, then the projection onto $\text{span}\{u\}$ is

$$\frac{uu^*}{u^*u}x.$$

An elementary reflector has the form

$$R = I - 2\hat{u}\hat{u}^* = I - \frac{2uu^*}{u^*u}, \quad u \neq 0,$$

and satisfies

$$R = R^* = R^{-1}, \quad R^2 = I.$$

So elementary reflectors are unitary, hermitian, and involutory.

Exercise 5.6.1

Brief restatement. Decide which of four displayed matrices are isometries.

An isometry preserves Euclidean length. For square real matrices this means orthogonal, and for square complex matrices this means unitary.

Part (a)

The matrix is

$$A = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}.$$

It is convenient to check the rows:

$$\begin{aligned} r_1 &= [1/\sqrt{2} \quad -1/\sqrt{2} \quad 0], \\ r_2 &= [1/\sqrt{6} \quad 1/\sqrt{6} \quad -2/\sqrt{6}], \\ r_3 &= [1/\sqrt{3} \quad 1/\sqrt{3} \quad 1/\sqrt{3}]. \end{aligned}$$

Their squared norms are

$$\begin{aligned} \|r_1\|^2 &= \frac{1}{2} + \frac{1}{2} = 1, \\ \|r_2\|^2 &= \frac{1}{6} + \frac{1}{6} + \frac{4}{6} = 1, \\ \|r_3\|^2 &= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1. \end{aligned}$$

The pairwise inner products are

$$r_1 r_2^T = \frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}} + 0 = 0,$$

$$r_1 r_3^T = \frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} + 0 = 0,$$

and

$$r_2 r_3^T = \frac{1}{\sqrt{18}} + \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} = 0.$$

Thus the rows are an orthonormal basis of $\mathbb{R}^3$. Therefore

$$AA^T = I.$$

For a square matrix, orthonormal rows are equivalent to orthonormal columns, so A is orthogonal.

Part (a) is an isometry.

Part (b)

The matrix is

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Check the column lengths:

$$B_{*1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \|B_{*1}\|^2 = 1^2 + 1^2 + 0^2 = 2.$$

The first column does not have length 1, so the columns are not orthonormal. Equivalently,

$$\|Be_1\| = \|B_{*1}\| = \sqrt{2} \neq 1 = \|e_1\|.$$

Thus B does not preserve all lengths.

Part (b) is not an isometry.

Part (c)

The matrix is the permutation matrix

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Every row contains exactly one 1, every column contains exactly one 1, and all other entries are 0. Therefore the rows and columns are orthonormal standard basis vectors in a different order. Hence

$$C^T C = I.$$

Part (c) is an isometry.

Part (d)

The matrix is

$$D = \begin{bmatrix} e^{i\theta_1} & 0 & \cdots & 0 \\ 0 & e^{i\theta_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{i\theta_n} \end{bmatrix}.$$

Because

$$|e^{i\theta_j}| = 1$$

for every real θ_j , each column has length 1, and different columns are orthogonal because the matrix is diagonal.

Equivalently,

$$D^*D = \text{diag}(e^{-i\theta_1}, \dots, e^{-i\theta_n}) \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) = I.$$

Thus D is unitary.

Part (d) is an isometry.

Final answer

The isometries are (a), (c), and (d).

Exercise 5.6.2

Brief restatement. Decide whether

$$U = \begin{bmatrix} (1+i)/\sqrt{3} & (1+i)/\sqrt{6} \\ i/\sqrt{3} & -2i/\sqrt{6} \end{bmatrix}$$

is unitary.

A matrix is unitary if its columns form an orthonormal set. Let

$$u_1 = \begin{bmatrix} (1+i)/\sqrt{3} \\ i/\sqrt{3} \end{bmatrix}, \quad u_2 = \begin{bmatrix} (1+i)/\sqrt{6} \\ -2i/\sqrt{6} \end{bmatrix}.$$

First compute the squared norms. Since $|1+i|^2 = 2$ and $|i|^2 = 1$,

$$\|u_1\|^2 = \frac{|1+i|^2}{3} + \frac{|i|^2}{3} = \frac{2}{3} + \frac{1}{3} = 1.$$

Similarly,

$$\|u_2\|^2 = \frac{|1+i|^2}{6} + \frac{|-2i|^2}{6} = \frac{2}{6} + \frac{4}{6} = 1.$$

Now compute the inner product $u_1^* u_2$:

$$\begin{aligned}
 u_1^* u_2 &= \frac{\overline{1+i}(1+i)}{\sqrt{3}\sqrt{6}} + \frac{\bar{i}(-2i)}{\sqrt{3}\sqrt{6}} \\
 &= \frac{(1-i)(1+i)}{\sqrt{18}} + \frac{(-i)(-2i)}{\sqrt{18}} \\
 &= \frac{2}{\sqrt{18}} + \frac{2i^2}{\sqrt{18}} \\
 &= \frac{2}{\sqrt{18}} - \frac{2}{\sqrt{18}} \\
 &= 0.
 \end{aligned}$$

The columns are orthonormal, so $U^* U = I$.

U is unitary.

Exercise 5.6.3

Brief restatement. Count diagonal matrices that are also orthogonal or unitary.

Part (a)

Let

$$D = \text{diag}(d_1, d_2, d_3)$$

be real and diagonal. Orthogonality means

$$D^T D = I.$$

Since $D^T = D$, this becomes

$$D^2 = \text{diag}(d_1^2, d_2^2, d_3^2) = I.$$

Therefore each diagonal entry satisfies

$$d_j^2 = 1,$$

so

$$d_j = \pm 1.$$

There are 2 choices for each of the 3 diagonal entries.

$$2^3 = 8$$

Part (b)

The same argument works for $n \times n$ real diagonal matrices. Each diagonal entry must independently be $+1$ or -1 .

$$2^n$$

Part (c)

Let

$$D = \text{diag}(z_1, \dots, z_n)$$

be complex diagonal. Unitarity means

$$D^* D = I.$$

Now

$$D^* D = \text{diag}(\bar{z}_1, \dots, \bar{z}_n) \text{diag}(z_1, \dots, z_n) = \text{diag}(|z_1|^2, \dots, |z_n|^2).$$

Therefore

$$|z_j|^2 = 1 \iff |z_j| = 1.$$

So each z_j may be any complex number on the unit circle:

$$z_j = e^{i\theta_j}, \quad \theta_j \in \mathbb{R}.$$

Thus the diagonal unitary matrices are exactly

$$D = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$$

and there are infinitely many of them.

Infinitely many; indeed, one for each point of the n -torus $(S^1)^n$.

Exercise 5.6.4

Brief restatement. Find conditions on real α, β that make two displayed matrices orthogonal/unitary.

Part (a)

The real matrix is

$$P = \begin{bmatrix} \alpha + \beta & \beta - \alpha \\ \alpha - \beta & \beta + \alpha \end{bmatrix}.$$

Let its columns be

$$p_1 = \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}, \quad p_2 = \begin{bmatrix} \beta - \alpha \\ \beta + \alpha \end{bmatrix}.$$

Compute the squared length of p_1 :

$$\begin{aligned} \|p_1\|^2 &= (\alpha + \beta)^2 + (\alpha - \beta)^2 \\ &= (\alpha^2 + 2\alpha\beta + \beta^2) + (\alpha^2 - 2\alpha\beta + \beta^2) \\ &= 2\alpha^2 + 2\beta^2 \\ &= 2(\alpha^2 + \beta^2). \end{aligned}$$

Similarly,

$$\|p_2\|^2 = (\beta - \alpha)^2 + (\beta + \alpha)^2 = 2(\alpha^2 + \beta^2).$$

Now compute the dot product:

$$\begin{aligned} p_1^T p_2 &= (\alpha + \beta)(\beta - \alpha) + (\alpha - \beta)(\beta + \alpha) \\ &= (\beta^2 - \alpha^2) + (\alpha^2 - \beta^2) \\ &= 0. \end{aligned}$$

Thus the two columns are always orthogonal. They are orthonormal exactly when their squared length is 1:

$$2(\alpha^2 + \beta^2) = 1.$$

Therefore

$$\boxed{\alpha^2 + \beta^2 = \frac{1}{2}}$$

is the necessary and sufficient condition.

Part (b)

The complex matrix is

$$U = \begin{bmatrix} 0 & \alpha & 0 & i\beta \\ \alpha & 0 & i\beta & 0 \\ 0 & i\beta & 0 & \alpha \\ i\beta & 0 & \alpha & 0 \end{bmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

Its columns are

$$u_1 = \begin{bmatrix} 0 \\ \alpha \\ 0 \\ i\beta \end{bmatrix}, \quad u_2 = \begin{bmatrix} \alpha \\ 0 \\ i\beta \\ 0 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 0 \\ i\beta \\ 0 \\ \alpha \end{bmatrix}, \quad u_4 = \begin{bmatrix} i\beta \\ 0 \\ \alpha \\ 0 \end{bmatrix}.$$

Each column has squared length

$$\alpha^2 + |i\beta|^2 = \alpha^2 + \beta^2.$$

The columns are pairwise orthogonal. Most pairs have disjoint nonzero positions. The only nontrivial cancellations are of the following type:

$$u_1^* u_3 = \overline{\alpha}(i\beta) + \overline{i\beta}\alpha = \alpha i\beta + (-i\beta)\alpha = 0,$$

and similarly

$$u_2^* u_4 = 0.$$

Thus the columns are mutually orthogonal for every real α, β . They are orthonormal exactly when

$$\alpha^2 + \beta^2 = 1.$$

Therefore

$$\boxed{\alpha^2 + \beta^2 = 1}$$

is the necessary and sufficient condition.

Exercise 5.6.5

Brief restatement. Show what happens to unitary/orthogonal matrices under products, sums, and block diagonal combinations.

The same algebra proves the unitary and orthogonal cases. I give the unitary proof; in the real orthogonal case replace $(\cdot)^*$ by $(\cdot)^T$.

Part (a)

Suppose U and V are $n \times n$ unitary matrices. Then

$$U^*U = I, \quad V^*V = I.$$

Compute:

$$\begin{aligned}(UV)^*(UV) &= V^*U^*UV \\ &= V^*(U^*U)V \\ &= V^*IV \\ &= V^*V \\ &= I.\end{aligned}$$

Hence UV is unitary.

UV is unitary.

In the real case,

$$(PQ)^T(PQ) = Q^T P^T PQ = Q^T Q = I,$$

so the product of orthogonal matrices is orthogonal.

Part (b)

The sum of two unitary matrices need not be unitary.

A simple counterexample is

$$U = I, \quad V = I.$$

Both are unitary, but

$$U + V = 2I.$$

Then

$$(2I)^*(2I) = 4I \neq I.$$

Thus $2I$ is not unitary.

Another counterexample is $U = I$, $V = -I$, for which $U + V = 0$, also not unitary.

$U + V$ need not be unitary.

Part (c)

Let $U \in \mathbb{C}^{n \times n}$ and $V \in \mathbb{C}^{m \times m}$ be unitary. Consider the block diagonal matrix

$$W = \begin{bmatrix} U & 0 \\ 0 & V \end{bmatrix}.$$

Then

$$W^* = \begin{bmatrix} U^* & 0 \\ 0 & V^* \end{bmatrix}.$$

Therefore

$$\begin{aligned} W^*W &= \begin{bmatrix} U^* & 0 \\ 0 & V^* \end{bmatrix} \begin{bmatrix} U & 0 \\ 0 & V \end{bmatrix} \\ &= \begin{bmatrix} U^*U & 0 \\ 0 & V^*V \end{bmatrix} \\ &= \begin{bmatrix} I_n & 0 \\ 0 & I_m \end{bmatrix} \\ &= I_{n+m}. \end{aligned}$$

Thus W is unitary. The real orthogonal case is identical with transposes.

$\begin{bmatrix} U & 0 \\ 0 & V \end{bmatrix}$ is unitary/orthogonal whenever U, V are.

Exercise 5.6.6

Brief restatement. Prove the Cayley transform

$$U = (I - A)(I + A)^{-1} = (I + A)^{-1}(I - A)$$

is unitary when A is skew hermitian. The real skew-symmetric case is analogous.

Assume

$$A^* = -A.$$

Step 1: Show that $I + A$ is invertible

It is enough to prove that

$$\mathcal{N}(I + A) = \{0\}.$$

Let $x \in \mathcal{N}(I + A)$. Then

$$(I + A)x = 0.$$

Take the inner product with x :

$$x^*(I + A)x = 0.$$

Expanding gives

$$x^*x + x^*Ax = 0.$$

Now x^*Ax is purely imaginary, because

$$\overline{x^*Ax} = x^*A^*x = x^*(-A)x = -x^*Ax.$$

Thus

$$\operatorname{Re}(x^*Ax) = 0.$$

Taking real parts in

$$x^*x + x^*Ax = 0$$

gives

$$\|x\|^2 = 0.$$

Therefore $x = 0$. Hence

$$\mathcal{N}(I + A) = \{0\}.$$

Since $I + A$ is square, it is invertible.

The same argument applied to $-A$, which is also skew hermitian, shows that $I - A$ is invertible too.

Step 2: Show the two displayed formulas agree

The matrices $I - A$ and $I + A$ commute:

$$(I - A)(I + A) = I - A^2 = (I + A)(I - A).$$

Since $I + A$ is invertible and commutes with $I - A$, its inverse also commutes with $I - A$. Therefore

$$(I - A)(I + A)^{-1} = (I + A)^{-1}(I - A).$$

Step 3: Prove unitarity

Let

$$U = (I - A)(I + A)^{-1}.$$

Then

$$\begin{aligned} U^* &= [(I - A)(I + A)^{-1}]^* \\ &= [(I + A)^{-1}]^* (I - A)^* \\ &= [(I + A)^*]^{-1} (I - A)^*. \end{aligned}$$

Because $A^* = -A$,

$$(I + A)^* = I - A, \quad (I - A)^* = I + A.$$

Thus

$$U^* = (I - A)^{-1}(I + A).$$

Now compute:

$$U^*U = (I - A)^{-1}(I + A)(I - A)(I + A)^{-1}.$$

Since $I + A$ commutes with $I - A$,

$$(I + A)(I - A) = (I - A)(I + A).$$

Therefore

$$\begin{aligned} U^*U &= (I - A)^{-1}(I - A)(I + A)(I + A)^{-1} \\ &= I. \end{aligned}$$

Hence U has orthonormal columns, so U is unitary.

U is unitary.

Real skew-symmetric case

If A is real and $A^T = -A$, replace every conjugate transpose $(\cdot)^*$ by transpose $(\cdot)^T$. The same proof gives

$$U^T U = I,$$

so U is orthogonal.

For real skew-symmetric A , the Cayley transform is orthogonal.

Exercise 5.6.7

Brief restatement. Determine whether certain block matrices built from elementary reflectors are themselves elementary reflectors.

Use the normalized form of an elementary reflector:

$$R = I - 2uu^*, \quad \|u\| = 1.$$

Part (a)

Suppose

$$R = I - 2uu^*$$

is an elementary reflector. Consider

$$M = \begin{bmatrix} I & 0 \\ 0 & R \end{bmatrix}.$$

Let

$$w = \begin{bmatrix} 0 \\ u \end{bmatrix}.$$

Then

$$\|w\|^2 = 0^*0 + u^*u = 1.$$

Also

$$ww^* = \begin{bmatrix} 0 \\ u \end{bmatrix} \begin{bmatrix} 0^* & u^* \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & uu^* \end{bmatrix}.$$

Therefore

$$I - 2ww^* = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - 2 \begin{bmatrix} 0 & 0 \\ 0 & uu^* \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I - 2uu^* \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & R \end{bmatrix}.$$

So the block matrix is an elementary reflector.

$\begin{bmatrix} I & 0 \\ 0 & R \end{bmatrix}$ is an elementary reflector.

Part (b)

Let

$$R = I - 2uu^*, \quad S = I - 2vv^*, \quad \|u\| = \|v\| = 1.$$

Then

$$\begin{bmatrix} R & 0 \\ 0 & S \end{bmatrix} = I - 2 \begin{bmatrix} uu^* & 0 \\ 0 & vv^* \end{bmatrix}.$$

If this were an elementary reflector, then for some unit vector w ,

$$\begin{bmatrix} R & 0 \\ 0 & S \end{bmatrix} = I - 2ww^*.$$

Thus

$$ww^* = \begin{bmatrix} uu^* & 0 \\ 0 & vv^* \end{bmatrix}.$$

But ww^* has rank 1, since all of its columns are scalar multiples of w . On the other hand,

$$\begin{bmatrix} uu^* & 0 \\ 0 & vv^* \end{bmatrix}$$

has rank 2, because uu^* and vv^* each have rank 1 and they act on separate coordinate blocks.

This is impossible. Therefore the block matrix is not a single elementary reflector.

$$\begin{bmatrix} R & 0 \\ 0 & S \end{bmatrix} \text{ is not an elementary reflector.}$$

It is still unitary, hermitian, and involutory; it is just not elementary.

Exercise 5.6.8

Brief restatement. Show that unitary transformations preserve inner products and orthogonal transformations preserve angles.

Part (a)

Let U be unitary, and define

$$u = Ux, \quad v = Uy.$$

Then

$$\begin{aligned} u^*v &= (Ux)^*(Uy) \\ &= x^*U^*Uy \\ &= x^*Iy \\ &= x^*y. \end{aligned}$$

So the standard inner product is unchanged by multiplication by a unitary matrix.

$$u^*v = x^*y.$$

Part (b)

Assume $x, y \neq 0$, since the angle with the zero vector is not defined. Let P be orthogonal and set

$$u = Px, \quad v = Py.$$

Because P is real orthogonal,

$$u^T v = (Px)^T (Py) = x^T P^T P y = x^T y.$$

Also,

$$\|u\| = \|Px\| = \|x\|, \quad \|v\| = \|Py\| = \|y\|.$$

Therefore

$$\begin{aligned} \cos \theta_{u,v} &= \frac{u^T v}{\|u\| \|v\|} \\ &= \frac{x^T y}{\|x\| \|y\|} \\ &= \cos \theta_{x,y}. \end{aligned}$$

$\cos \theta_{u,v} = \cos \theta_{x,y}.$

Thus orthogonal transformations preserve angles.

Exercise 5.6.9

Brief restatement. Let $U \in \mathbb{C}^{m \times r}$ have orthonormal columns, let $V \in \mathbb{C}^{k \times n}$ have orthonormal rows, and let $A \in \mathbb{C}^{r \times k}$. Compute norms of U, V and prove invariance of the 2-norm and Frobenius norm under multiplication by U, V .

The assumptions mean

$$U^* U = I_r, \quad V V^* = I_k.$$

Part (a): Compute $\|U\|_2$ and $\|V\|_2$

For every $x \in \mathbb{C}^r$,

$$\|Ux\|^2 = (Ux)^*(Ux) = x^* U^* U x = x^* x = \|x\|^2.$$

Thus multiplication by U preserves the lengths of vectors in $\mathbb{C}^r$. Hence

$\|U\|_2 = 1.$

Now consider V . Since $V V^* = I_k$, the matrix $V^* \in \mathbb{C}^{n \times k}$ has orthonormal columns. Therefore for every $z \in \mathbb{C}^k$,

$$\|V^* z\| = \|z\|.$$

This implies $\|V^*\|_2 = 1$. Also, the matrix 2-norm is unchanged by taking the conjugate transpose:

$$\|V\|_2 = \|V^*\|_2.$$

Therefore

$\|V\|_2 = 1.$

One can also see the upper bound directly: $V^* V$ is an orthogonal projector, so $\|Vy\| \leq \|y\|$ for every y , while equality occurs for $y = V^* z$ with $\|z\| = 1$.

Part (a): Compute $\|U\|_F$ and $\|V\|_F$

The Frobenius norm is the square root of the sum of the squared moduli of all entries. This is also the square root of the sum of the squared column lengths.

The matrix U has r orthonormal columns, so each column has squared length 1. Hence

$$\|U\|_F^2 = \sum_{j=1}^r \|U_{*j}\|^2 = \sum_{j=1}^r 1 = r.$$

Therefore

$$\boxed{\|U\|_F = \sqrt{r}.$$

Similarly, V has k orthonormal rows, so the sum of squared entries row-by-row is

$$\|V\|_F^2 = \sum_{i=1}^k \|V_{i*}\|^2 = \sum_{i=1}^k 1 = k.$$

Therefore

$$\boxed{\|V\|_F = \sqrt{k}.$$

Part (b): Prove $\|UAV\|_2 = \|A\|_2$

First prove

$$\|UA\|_2 = \|A\|_2.$$

For every $x \in \mathbb{C}^k$,

$$\|UAx\| = \|Ax\|,$$

because $Ax \in \mathbb{C}^r$ and U preserves lengths on $\mathbb{C}^r$. Taking the maximum over all $\|x\| = 1$ gives

$$\|UA\|_2 = \|A\|_2.$$

Next prove

$$\|AV\|_2 = \|A\|_2.$$

For every $y \in \mathbb{C}^n$,

$$\|AVy\| \leq \|A\|_2 \|Vy\| \leq \|A\|_2 \|y\|.$$

Thus

$$\|AV\|_2 \leq \|A\|_2.$$

For the reverse inequality, choose any $z \in \mathbb{C}^k$ with $\|z\| = 1$. Set

$$y = V^*z.$$

Then

$$\|y\| = \|V^*z\| = \|z\| = 1,$$

and

$$Vy = VV^*z = z.$$

Therefore

$$\|AVy\| = \|Az\|.$$

Taking the maximum over all unit z gives

$$\|AV\|_2 \geq \|A\|_2.$$

So

$$\|AV\|_2 = \|A\|_2.$$

Combining the two results,

$$\|UAV\|_2 = \|U(AV)\|_2 = \|AV\|_2 = \|A\|_2.$$

$$\boxed{\|UAV\|_2 = \|A\|_2.}$$

Part (c): Prove $\|UAV\|_F = \|A\|_F$

First,

$$UA = [UA_{*1} \quad UA_{*2} \quad \cdots \quad UA_{*k}].$$

Since U preserves vector lengths,

$$\|UA_{*j}\| = \|A_{*j}\|$$

for every column A_{*j} . Hence

$$\begin{aligned} \|UA\|_F^2 &= \sum_{j=1}^k \|UA_{*j}\|^2 \\ &= \sum_{j=1}^k \|A_{*j}\|^2 \\ &= \|A\|_F^2. \end{aligned}$$

So

$$\|UA\|_F = \|A\|_F.$$

Now consider AV . Let $a_i = A_{i*}$ be the i -th row of A . Then the i -th row of AV is

$$a_i V.$$

Its norm satisfies

$$\|a_i V\| = \|(a_i V)^*\| = \|V^* a_i^*\| = \|a_i^*\| = \|a_i\|,$$

because V^* has orthonormal columns and therefore preserves lengths on $\mathbb{C}^k$.

Thus

$$\begin{aligned} \|AV\|_F^2 &= \sum_{i=1}^r \|a_i V\|^2 \\ &= \sum_{i=1}^r \|a_i\|^2 \\ &= \|A\|_F^2. \end{aligned}$$

So

$$\|AV\|_F = \|A\|_F.$$

Combining,

$$\|UAV\|_F = \|U(AV)\|_F = \|AV\|_F = \|A\|_F.$$

$$\boxed{\|UAV\|_F = \|A\|_F.}$$

Exercise 5.6.10

Brief restatement. For

$$u = \begin{bmatrix} -2 \\ 1 \\ 3 \\ -1 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 4 \\ 0 \\ -1 \end{bmatrix},$$

compute four orthogonal projections.

First compute the inner products:

$$u^T u = (-2)^2 + 1^2 + 3^2 + (-1)^2 = 4 + 1 + 9 + 1 = 15,$$

$$v^T v = 1^2 + 4^2 + 0^2 + (-1)^2 = 1 + 16 + 0 + 1 = 18,$$

and

$$u^T v = (-2)(1) + (1)(4) + (3)(0) + (-1)(-1) = -2 + 4 + 0 + 1 = 3.$$

Part (a): Projection of u onto $\text{span}\{v\}$

The projection formula is

$$\text{proj}_{\text{span}\{v\}}(u) = \frac{v^T u}{v^T v} v.$$

Thus

$$\text{proj}_{\text{span}\{v\}}(u) = \frac{3}{18} v = \frac{1}{6} v = \begin{bmatrix} 1/6 \\ 2/3 \\ 0 \\ -1/6 \end{bmatrix}.$$

$$\boxed{\text{proj}_{\text{span}\{v\}}(u) = \frac{1}{6} \begin{bmatrix} 1 \\ 4 \\ 0 \\ -1 \end{bmatrix}}$$

Part (b): Projection of v onto $\text{span}\{u\}$

$$\text{proj}_{\text{span}\{u\}}(v) = \frac{u^T v}{u^T u} u = \frac{3}{15} u = \frac{1}{5} u.$$

Therefore

$$\text{proj}_{\text{span}\{u\}}(v) = \begin{bmatrix} -2/5 \\ 1/5 \\ 3/5 \\ -1/5 \end{bmatrix}.$$

$$\text{proj}_{\text{span}\{u\}}(v) = \frac{1}{5} \begin{bmatrix} -2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

Part (c): Projection of u onto $v^\perp$

The vector u splits orthogonally as

$$u = \text{proj}_{\text{span}\{v\}}(u) + \text{proj}_{v^\perp}(u).$$

Hence

$$\text{proj}_{v^\perp}(u) = u - \text{proj}_{\text{span}\{v\}}(u).$$

Using part (a),

$$\begin{aligned} \text{proj}_{v^\perp}(u) &= \begin{bmatrix} -2 \\ 1 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 1/6 \\ 2/3 \\ 0 \\ -1/6 \end{bmatrix} \\ &= \begin{bmatrix} -13/6 \\ 1/3 \\ 3 \\ -5/6 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} -13 \\ 2 \\ 18 \\ -5 \end{bmatrix}. \end{aligned}$$

Check:

$$v^T \text{proj}_{v^\perp}(u) = 1 \left(-\frac{13}{6} \right) + 4 \left(\frac{1}{3} \right) + 0(3) + (-1) \left(-\frac{5}{6} \right) = -\frac{13}{6} + \frac{8}{6} + \frac{5}{6} = 0.$$

$$\text{proj}_{v^\perp}(u) = \frac{1}{6} \begin{bmatrix} -13 \\ 2 \\ 18 \\ -5 \end{bmatrix}$$

Part (d): Projection of v onto $u^\perp$

Similarly,

$$\text{proj}_{u^\perp}(v) = v - \text{proj}_{\text{span}\{u\}}(v).$$

Using part (b),

$$\begin{aligned}\text{proj}_{u^\perp}(v) &= \begin{bmatrix} 1 \\ 4 \\ 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 1/5 \\ 3/5 \\ -1/5 \end{bmatrix} \\ &= \begin{bmatrix} 7/5 \\ 19/5 \\ -3/5 \\ -4/5 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 7 \\ 19 \\ -3 \\ -4 \end{bmatrix}.\end{aligned}$$

Check:

$$u^T \text{proj}_{u^\perp}(v) = (-2)\frac{7}{5} + 1\frac{19}{5} + 3\left(-\frac{3}{5}\right) + (-1)\left(-\frac{4}{5}\right) = \frac{-14 + 19 - 9 + 4}{5} = 0.$$

$$\text{proj}_{u^\perp}(v) = \frac{1}{5} \begin{bmatrix} 7 \\ 19 \\ -3 \\ -4 \end{bmatrix}$$

Exercise 5.6.11

Brief restatement. For an elementary orthogonal projector

$$Q = I - uu^*, \quad \|u\| = 1,$$

prove singularity and compute the rank.

Part (a)

Compute Qu :

$$Qu = (I - uu^*)u = u - u(u^*u).$$

Since $\|u\| = 1$,

$$u^*u = 1.$$

Therefore

$$Qu = u - u = 0.$$

Also $u \neq 0$, since $\|u\| = 1$. Thus Q has a nonzero vector in its nullspace:

$$u \in \mathcal{N}(Q), \quad u \neq 0.$$

Therefore Q is singular.

$Q \text{ is singular.}$

Part (b)

Now suppose $Q \in \mathbb{C}^{n \times n}$. We already know

$$\text{span}\{u\} \subseteq \mathcal{N}(Q).$$

Conversely, suppose $x \in \mathcal{N}(Q)$. Then

$$Qx = 0.$$

So

$$(I - uu^*)x = 0.$$

Hence

$$x = uu^*x.$$

But u^*x is a scalar, so

$$x = (u^*x)u.$$

Therefore

$$x \in \text{span}\{u\}.$$

Thus

$$\mathcal{N}(Q) = \text{span}\{u\}.$$

Because $u \neq 0$,

$$\dim \mathcal{N}(Q) = 1.$$

By the rank-nullity theorem,

$$\text{rank}(Q) + \dim \mathcal{N}(Q) = n.$$

Therefore

$$\text{rank}(Q) + 1 = n,$$

and hence

$$\boxed{\text{rank}(Q) = n - 1.}$$

Exercise 5.6.12

Brief restatement. Let $\|u\| = 1$, and let p be the orthogonal projection of x onto $\text{span}\{u\}$. Prove $\|p\| \leq \|x\|$, with equality exactly when x is a scalar multiple of u .

Because $\|u\| = 1$, the projection is

$$p = uu^*x.$$

Equivalently,

$$p = (u^*x)u.$$

Thus

$$\|p\| = \|(u^*x)u\| = |u^*x|\|u\| = |u^*x|.$$

By the Cauchy–Schwarz inequality,

$$|u^*x| \leq \|u\|\|x\| = \|x\|.$$

Therefore

$$\boxed{\|p\| \leq \|x\|.}$$

Now determine when equality holds.

One way is to use the orthogonal decomposition

$$x = p + (x - p),$$

where

$$p \in \text{span}\{u\}, \quad x - p \in u^\perp.$$

So

$$p \perp (x - p).$$

By the Pythagorean theorem,

$$\|x\|^2 = \|p\|^2 + \|x - p\|^2.$$

Thus

$$\|p\| = \|x\|$$

if and only if

$$\|x - p\| = 0,$$

which holds if and only if

$$x = p.$$

But $p \in \text{span}\{u\}$, so this is equivalent to saying that x is a scalar multiple of u .

Conversely, if $x = \alpha u$, then

$$p = uu^*(\alpha u) = \alpha u(u^*u) = \alpha u = x,$$

so $\|p\| = \|x\|$.

$\|p\| = \|x\| \iff x \in \text{span}\{u\}.$

Exercise 5.6.13

Brief restatement. For

$$x = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix},$$

construct an elementary reflector mapping x onto the x -axis, verify its properties, and use it to extend x to an orthonormal basis of $\mathbb{R}^3$.

First note that

$$\|x\|^2 = \frac{1}{9}(1^2 + (-2)^2 + (-2)^2) = \frac{1}{9}(1 + 4 + 4) = 1.$$

Part (a)

We want $Rx = e_1$, where

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

For a Householder reflector, choose a vector parallel to $x - e_1$:

$$x - e_1 = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} = -\frac{2}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

The scalar factor is irrelevant in

$$R = I - \frac{2ww^T}{w^Tw},$$

so take

$$w = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Then

$$w^Tw = 3, \quad ww^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Therefore

$$R = I - \frac{2}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}.$$

Now verify that it sends x to e_1 :

$$\begin{aligned} Rx &= \frac{1}{3} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 1 + 4 + 4 \\ -2 - 2 + 4 \\ -2 + 4 - 2 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = e_1. \end{aligned}$$

Thus Rx lies on the x -axis.

$$R = \frac{1}{3} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

Part (b)

Let

$$J = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Then

$$R = I - \frac{2}{3}J.$$

The matrix J is symmetric, so R is symmetric:

$$R^T = R.$$

Also,

$$J^2 = 3J,$$

because each entry of J^2 is the sum $1 + 1 + 1 = 3$.

Therefore

$$\begin{aligned} R^2 &= \left(I - \frac{2}{3}J\right)^2 \\ &= I - \frac{4}{3}J + \frac{4}{9}J^2 \\ &= I - \frac{4}{3}J + \frac{4}{9}(3J) \\ &= I - \frac{4}{3}J + \frac{4}{3}J \\ &= I. \end{aligned}$$

Thus R is involutory:

$$\boxed{R^2 = I.}$$

Since $R^T = R$, we also have

$$R^T R = R^2 = I.$$

Therefore R is orthogonal:

$$\boxed{R^T R = I.}$$

So R is symmetric, orthogonal, and involutory.

Part (c)

Because R is orthogonal, its columns form an orthonormal basis of $\mathbb{R}^3$. The first column is

$$R_{*1} = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} = x.$$

Therefore the columns of R extend x to an orthonormal basis:

$$\boxed{\left\{ \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}, \frac{1}{3} \begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix}, \frac{1}{3} \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \right\}}.$$

Exercise 5.6.14

Brief restatement. Let

$$R = I - 2uu^*, \quad \|u\| = 1.$$

If $Rx = x$, prove $x \perp u$. Interpret geometrically in $\mathbb{R}^3$.

Assume

$$Rx = x.$$

Substitute the formula for R :

$$(I - 2uu^*)x = x.$$

Expand:

$$x - 2uu^*x = x.$$

Subtract x from both sides:

$$-2uu^*x = 0.$$

Thus

$$uu^*x = 0.$$

Since u^*x is a scalar,

$$uu^*x = (u^*x)u.$$

Because $\|u\| = 1$, we know $u \neq 0$. Therefore

$$(u^*x)u = 0$$

forces

$$u^*x = 0.$$

This is exactly the statement

$$x \perp u.$$

$$\boxed{Rx = x \implies x \in u^\perp.}$$

Geometric interpretation in $\mathbb{R}^3$

In $\mathbb{R}^3$, the set

$$u^\perp = \{x \in \mathbb{R}^3 : u^T x = 0\}$$

is the plane through the origin perpendicular to u . The reflector

$$R = I - 2uu^T$$

leaves every vector in this plane fixed and reverses the component in the u -direction. If

$$x = x_\parallel + x_\perp, \quad x_\parallel \in \text{span}\{u\}, \quad x_\perp \in u^\perp,$$

then

$$Rx = -x_\parallel + x_\perp.$$

Thus $Rx = x$ can happen only when $x_\parallel = 0$, i.e. when $x \in u^\perp$.

Exercise 5.6.15

Brief restatement. Given real vectors $x, y \in \mathbb{R}^n$ with $\|x\| = \|y\|$ and $x \neq y$, construct an elementary reflector R such that $Rx = y$.

Since $x \neq y$, the vector

$$u = x - y$$

is nonzero. Define the reflector

$$R = I - \frac{2uu^T}{u^T u}.$$

We claim that

$$Rx = y.$$

Compute:

$$Rx = x - \frac{2u(u^T x)}{u^T u}.$$

Since $u = x - y$,

$$u^T x = (x - y)^T x = x^T x - y^T x = \|x\|^2 - x^T y.$$

Also,

$$\begin{aligned} u^T u &= (x - y)^T (x - y) \\ &= x^T x - 2x^T y + y^T y \\ &= \|x\|^2 - 2x^T y + \|y\|^2. \end{aligned}$$

Because $\|x\| = \|y\|$,

$$u^T u = 2\|x\|^2 - 2x^T y = 2(\|x\|^2 - x^T y).$$

Therefore

$$\frac{2u^T x}{u^T u} = \frac{2(\|x\|^2 - x^T y)}{2(\|x\|^2 - x^T y)} = 1.$$

The denominator is nonzero because $u = x - y \neq 0$. Hence

$$Rx = x - u = x - (x - y) = y.$$

Thus

$$R = I - \frac{2(x - y)(x - y)^T}{(x - y)^T (x - y)}$$

is an elementary reflector satisfying

$$Rx = y.$$

Geometric note

The reflecting hyperplane is perpendicular to $x - y$. Since $\|x\| = \|y\|$,

$$(x - y)^T (x + y) = \|x\|^2 - \|y\|^2 = 0,$$

so the vector $x + y$ lies in the reflecting hyperplane. Reflection across that hyperplane swaps x and y .

Exercise 5.6.16

Brief restatement. Given a unit vector

$$x = \begin{bmatrix} x_1 \\ \tilde{x} \end{bmatrix},$$

show that two displayed block matrices extend x to an orthonormal basis.

Part (a): Real case

Assume $x \in \mathbb{R}^n$, $\|x\| = 1$, and $x_1 \neq 1$. Let

$$z = \tilde{x}, \quad a = x_1.$$

Then

$$x = \begin{bmatrix} a \\ z \end{bmatrix}, \quad a^2 + z^T z = 1.$$

The matrix in the exercise is

$$P = \begin{bmatrix} a & z^T \\ z & I - \alpha z z^T \end{bmatrix}, \quad \alpha = \frac{1}{1-a}.$$

We show that P is an elementary reflector.

Set

$$w = x - e_1 = \begin{bmatrix} a-1 \\ z \end{bmatrix}.$$

Because $a \neq 1$, $w \neq 0$. Compute

$$\begin{aligned} w^T w &= (a-1)^2 + z^T z \\ &= (a-1)^2 + (1-a^2) \\ &= a^2 - 2a + 1 + 1 - a^2 \\ &= 2(1-a). \end{aligned}$$

Therefore

$$I - \frac{2ww^T}{w^T w} = I - \frac{ww^T}{1-a}.$$

Now compute this block-by-block. Since

$$ww^T = \begin{bmatrix} (a-1)^2 & (a-1)z^T \\ z(a-1) & zz^T \end{bmatrix},$$

we have

$$I - \frac{ww^T}{1-a} = \begin{bmatrix} 1 - \frac{(a-1)^2}{1-a} & -\frac{(a-1)z^T}{1-a} \\ -\frac{z(a-1)}{1-a} & I - \frac{zz^T}{1-a} \end{bmatrix}.$$

Because

$$\frac{(a-1)^2}{1-a} = 1-a, \quad -\frac{a-1}{1-a} = 1,$$

this becomes

$$\begin{bmatrix} a & z^T \\ z & I - \alpha z z^T \end{bmatrix} = P.$$

Thus

$$P = I - \frac{2ww^T}{w^T w}.$$

This is an elementary reflector, so it is orthogonal.

$$\boxed{P^T P = I.}$$

Also, its first column is exactly x , so its columns form an orthonormal basis containing x .

Part (b): Complex case

The displayed complex formula is

$$U = \begin{bmatrix} x_1 & \mu^2 \tilde{x}^* \\ \tilde{x} & \mu(I - \alpha \tilde{x} \tilde{x}^*) \end{bmatrix}, \quad \alpha = \frac{1}{1 - |x_1|}.$$

Important convention note. The proof requires μ to be the phase of x_1 :

$$x_1 = \mu |x_1|, \quad |\mu| = 1,$$

with $\mu = 1$ when $x_1 = 0$. This is the convention under which the displayed formula is correct. If one reads the earlier printed convention literally as $\mu = 1$ for every real x_1 , then the formula fails for negative real x_1 . For example, $x = (-1/2, \sqrt{3}/2)^T$ gives nonorthogonal columns. The intended phase convention is used below.

Let

$$a = x_1, \quad \rho = |a|, \quad z = \tilde{x},$$

so that

$$a = \mu \rho, \quad z^* z = 1 - \rho^2.$$

Assume $|x_1| \neq 1$, so $\rho \neq 1$. Define

$$w = x - \mu e_1 = \begin{bmatrix} a - \mu \\ z \end{bmatrix}.$$

Since $a = \mu \rho$,

$$a - \mu = \mu(\rho - 1) = -\mu(1 - \rho).$$

Therefore

$$\begin{aligned} w^* w &= |a - \mu|^2 + z^* z \\ &= (1 - \rho)^2 + (1 - \rho^2) \\ &= (1 - \rho)^2 + (1 - \rho)(1 + \rho) \\ &= 2(1 - \rho). \end{aligned}$$

Thus

$$\frac{2}{w^* w} = \frac{1}{1 - \rho} = \alpha.$$

Let

$$H = I - \frac{2ww^*}{w^* w} = I - \alpha ww^*.$$

This is an elementary reflector, so H is unitary. Since $|\mu| = 1$, the scalar multiple

$$\mu H$$

is also unitary:

$$(\mu H)^*(\mu H) = \bar{\mu}\mu H^*H = I.$$

Now verify that μH is exactly the displayed matrix.

Because

$$ww^* = \begin{bmatrix} |a - \mu|^2 & (a - \mu)z^* \\ z(\bar{a} - \bar{\mu}) & zz^* \end{bmatrix},$$

we get

$$H = \begin{bmatrix} 1 - \alpha|a - \mu|^2 & -\alpha(a - \mu)z^* \\ -\alpha z(\bar{a} - \bar{\mu}) & I - \alpha zz^* \end{bmatrix}.$$

Now compute the blocks of μH .

Top-left block:

$$\mu(1 - \alpha|a - \mu|^2) = \mu\left(1 - \frac{(1 - \rho)^2}{1 - \rho}\right) = \mu\rho = a = x_1.$$

Top-right block:

$$\mu[-\alpha(a - \mu)z^*] = \mu[-\alpha\mu(\rho - 1)z^*] = \mu(\mu z^*) = \mu^2 z^*.$$

Bottom-left block:

$$\mu[-\alpha z(\bar{a} - \bar{\mu})] = \mu[-\alpha z\bar{\mu}(\rho - 1)] = \mu(\bar{\mu}z) = z.$$

Bottom-right block:

$$\mu(I - \alpha zz^*).$$

Therefore

$$\mu H = \begin{bmatrix} x_1 & \mu^2 z^* \\ z & \mu(I - \alpha zz^*) \end{bmatrix} = U.$$

Since $U = \mu H$ and H is unitary,

$$\boxed{U^*U = I.}$$

The first column of U is x , so the columns of U form an orthonormal basis containing the original vector x .

Exercise 5.6.17

Brief restatement. Starting with

$$v_0 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix},$$

apply rotations: 45° counterclockwise about the x -axis, 90° clockwise about the y -axis, and 30° counterclockwise about the z -axis. Find v_3 and the orthogonal matrix Q with $Qv_0 = v_3$.

The rotation matrices from the section are

$$P_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix},$$

$$P_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix},$$

and

$$P_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The requested sequence is

$$Q = P_z(\pi/6)P_y(-\pi/2)P_x(\pi/4).$$

Step 1: Rotate about the x -axis

Since

$$\cos(\pi/4) = \sin(\pi/4) = \frac{\sqrt{2}}{2},$$

$$v_1 = P_x(\pi/4)v_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Thus

$$v_1 = \begin{bmatrix} 1 \\ \sqrt{2}/2 + \sqrt{2}/2 \\ \sqrt{2}/2 - \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{2} \\ 0 \end{bmatrix}.$$

Step 2: Rotate clockwise 90° about the y -axis

Clockwise 90° means counterclockwise angle $-\pi/2$. Hence

$$P_y(-\pi/2) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Therefore

$$v_2 = P_y(-\pi/2)v_1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ \sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}.$$

Step 3: Rotate about the z -axis

Since

$$\cos(\pi/6) = \frac{\sqrt{3}}{2}, \quad \sin(\pi/6) = \frac{1}{2},$$

$$P_z(\pi/6) = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then

$$v_3 = P_z(\pi/6)v_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}.$$

So

$$v_3 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{6}/2 \\ 1 \end{bmatrix}.$$

$$v_3 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{6}/2 \\ 1 \end{bmatrix}.$$

Compute Q

Now multiply

$$Q = P_z(\pi/6)P_y(-\pi/2)P_x(\pi/4).$$

The product is

$$Q = \frac{1}{4} \begin{bmatrix} 0 & -\sqrt{6} - \sqrt{2} & -\sqrt{6} + \sqrt{2} \\ 0 & \sqrt{6} - \sqrt{2} & -\sqrt{6} - \sqrt{2} \\ 4 & 0 & 0 \end{bmatrix}.$$

Check:

$$Qv_0 = \frac{1}{4} \begin{bmatrix} 0 & -\sqrt{6} - \sqrt{2} & -\sqrt{6} + \sqrt{2} \\ 0 & \sqrt{6} - \sqrt{2} & -\sqrt{6} - \sqrt{2} \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The components are

$$\frac{1}{4} [(-\sqrt{6} - \sqrt{2}) + (\sqrt{6} - \sqrt{2})] = -\frac{\sqrt{2}}{2},$$

$$\frac{1}{4} [(\sqrt{6} - \sqrt{2}) + (-\sqrt{6} - \sqrt{2})] = -\frac{\sqrt{2}}{2},$$

and

$$\frac{1}{4}(4) = 1.$$

Thus

$$Qv_0 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{6}/2 \\ 1 \end{bmatrix} = v_3.$$

Because Q is a product of orthogonal rotation matrices, Q is orthogonal.

$$Q = \frac{1}{4} \begin{bmatrix} 0 & -\sqrt{6} - \sqrt{2} & -\sqrt{6} + \sqrt{2} \\ 0 & \sqrt{6} - \sqrt{2} & -\sqrt{6} - \sqrt{2} \\ 4 & 0 & 0 \end{bmatrix}.$$

Exercise 5.6.18

Brief restatement. Decide whether the order of rotations in $\mathbb{R}^3$ matters.

Yes, the order generally matters. Matrix multiplication is not commutative, and rotation matrices around different axes usually do not commute.

Let

$$P_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\theta & -s_\theta \\ 0 & s_\theta & c_\theta \end{bmatrix}, \quad P_y(\phi) = \begin{bmatrix} c_\phi & 0 & s_\phi \\ 0 & 1 & 0 \\ -s_\phi & 0 & c_\phi \end{bmatrix},$$

where

$$c_\theta = \cos \theta, \quad s_\theta = \sin \theta, \quad c_\phi = \cos \phi, \quad s_\phi = \sin \phi.$$

If we first rotate around the x -axis and then around the y -axis, the resulting matrix is

$$P_y(\phi)P_x(\theta).$$

Compute:

$$P_y(\phi)P_x(\theta) = \begin{bmatrix} c_\phi & s_\phi s_\theta & s_\phi c_\theta \\ 0 & c_\theta & -s_\theta \\ -s_\phi & c_\phi s_\theta & c_\phi c_\theta \end{bmatrix}.$$

If we first rotate around the y -axis and then around the x -axis, the resulting matrix is

$$P_x(\theta)P_y(\phi).$$

Compute:

$$P_x(\theta)P_y(\phi) = \begin{bmatrix} c_\phi & 0 & s_\phi \\ s_\theta s_\phi & c_\theta & -s_\theta c_\phi \\ -c_\theta s_\phi & s_\theta & c_\theta c_\phi \end{bmatrix}.$$

These matrices are not generally equal. For example, their $(1, 2)$ -entries are

$$s_\phi s_\theta \quad \text{and} \quad 0.$$

These are different whenever

$$\sin \phi \sin \theta \neq 0.$$

A concrete vector example is even more direct. Let

$$\theta = \phi = \frac{\pi}{2}, \quad v = e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

First rotate about x , then about y :

$$P_x(\pi/2)e_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix},$$

and then

$$P_y(\pi/2) \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}.$$

Now reverse the order. First rotate about y :

$$P_y(\pi/2)e_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

and then rotate about x :

$$P_x(\pi/2) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The two final vectors are different:

$$\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The order of rotations in $\mathbb{R}^3$ generally matters.

Exercise 5.6.19

Brief restatement. For nonzero $u \in \mathbb{C}^n$, prove

$$\dim u^\perp = n - 1.$$

Define the linear functional

$$T : \mathbb{C}^n \rightarrow \mathbb{C}, \quad T(x) = u^*x.$$

Then

$$\mathcal{N}(T) = \{x \in \mathbb{C}^n : u^*x = 0\} = u^\perp.$$

Because $u \neq 0$,

$$T(u) = u^*u = \|u\|^2 \neq 0.$$

So T is not the zero transformation.

In fact, T maps onto all of $\mathbb{C}$. To see this, let $\alpha \in \mathbb{C}$. Set

$$x = \frac{\alpha}{\|u\|^2}u.$$

Then

$$T(x) = u^* \left(\frac{\alpha}{\|u\|^2}u \right) = \frac{\alpha}{\|u\|^2}u^*u = \alpha.$$

Thus

$$\mathcal{R}(T) = \mathbb{C},$$

so

$$\text{rank}(T) = 1.$$

By rank-nullity,

$$\dim \mathcal{N}(T) + \text{rank}(T) = n.$$

Therefore

$$\dim u^\perp + 1 = n,$$

and hence

$$\boxed{\dim u^\perp = n - 1.}$$

Exercise 5.6.20

Brief restatement. Show a one-to-one correspondence between involutions $A^2 = I$ and projectors $P^2 = P$ in $\mathbb{C}^{n \times n}$.

Let

$$\mathcal{I} = \{A \in \mathbb{C}^{n \times n} : A^2 = I\}$$

be the set of involutions, and let

$$\mathcal{P} = \{P \in \mathbb{C}^{n \times n} : P^2 = P\}$$

be the set of projectors.

From projectors to involutions

Given a projector P , define

$$A = I - 2P.$$

Then

$$\begin{aligned} A^2 &= (I - 2P)^2 \\ &= I - 4P + 4P^2. \end{aligned}$$

Since $P^2 = P$,

$$A^2 = I - 4P + 4P = I.$$

Thus A is an involution.

From involutions to projectors

Given an involution A , define

$$P = \frac{I - A}{2}.$$

Then

$$\begin{aligned} P^2 &= \left(\frac{I - A}{2} \right)^2 \\ &= \frac{1}{4}(I - 2A + A^2). \end{aligned}$$

Since $A^2 = I$,

$$P^2 = \frac{1}{4}(I - 2A + I) = \frac{1}{2}(I - A) = P.$$

Thus P is a projector.

Show the two maps are inverses

Start with a projector P , form

$$A = I - 2P,$$

and then form

$$\frac{I - A}{2} = \frac{I - (I - 2P)}{2} = \frac{2P}{2} = P.$$

Start with an involution A , form

$$P = \frac{I - A}{2},$$

and then form

$$I - 2P = I - 2\left(\frac{I - A}{2}\right) = I - (I - A) = A.$$

Thus the maps

$$P \mapsto I - 2P$$

and

$$A \mapsto \frac{I - A}{2}$$

are inverse to each other.

Therefore there is a one-to-one correspondence between projectors and involutions.

$$P^2 = P \iff A = I - 2P \text{ satisfies } A^2 = I,$$

with inverse

$$P = \frac{I - A}{2}.$$

Exercise 5.6.21

Brief restatement. Explain why, for a convex polytope viewed along the positive x -direction toward the yz -screen, a face with outward normal n is visible exactly when $n_1 > 0$.

In the setup from the section, the screen is the yz -plane, and the positive x -axis points toward the viewer. Therefore a face is visible from the viewer's perspective precisely when its outward normal has a component pointing toward the viewer.

Let

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

be the unit vector in the positive x -direction.

The outward normal vector n makes an angle θ with the positive x -axis. The face is front-facing exactly when this angle is acute:

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Equivalently,

$$\cos \theta > 0.$$

Using the dot product formula,

$$\cos \theta = \frac{n^T e_1}{\|n\| \|e_1\|}.$$

Since $\|n\| > 0$ and $\|e_1\| = 1$, the sign of $\cos \theta$ is the sign of $n^T e_1$. But

$$n^T e_1 = \begin{bmatrix} n_1 & n_2 & n_3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = n_1.$$

Therefore

$$\cos \theta > 0 \iff n_1 > 0.$$

Thus the outward side of the face is visible exactly when

$$\boxed{n_1 > 0.}$$

If $n_1 < 0$, the outward normal points away from the viewer, so the face is hidden on a convex polytope. If $n_1 = 0$, the face is edge-on relative to the viewer and has zero projected area on the screen.

Appendix: recurring checks

Checking orthogonality/unitarity

For a real square matrix P ,

$$P \text{ orthogonal} \iff P^T P = I.$$

For a complex square matrix U ,

$$U \text{ unitary} \iff U^* U = I.$$

Projection onto a one-dimensional subspace

For $u \neq 0$,

$$\text{proj}_{\text{span}\{u\}}(x) = \frac{uu^*}{u^*u}x = \frac{u^*x}{u^*u}u.$$

The projection onto $u^\perp$ is

$$x - \text{proj}_{\text{span}\{u\}}(x) = \left(I - \frac{uu^*}{u^*u} \right) x.$$

Elementary reflector

For $u \neq 0$,

$$R = I - \frac{2uu^*}{u^*u}$$

satisfies

$$R^* = R, \quad R^2 = I, \quad R^{-1} = R.$$